Realtime Voice Migration Analysis

OpenAI Realtime vs Gemini Live vs Retell+Gemini vs LiveKit — Strategic Comparison & Migration Plan

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Realtime Voice Stack Migration Analysis — JM Cafe Pilot

Date: 2026-05-05 | **Author**: SMB Voice Agent Researcher | **Status**: Strategic Decision Memo
Classification: BD / Engineering — Pre-Launch Architecture Review
Subject Pilot: JM Cafe (Portland, OR) — single-store, pre-commercial
Current Stack: Retell AI (voice infra) + Gemini 3.1 Flash Lite (LLM) + FastAPI Bridge + Loyverse POS + Twilio SMS pay link
Decision Surface: Stay on Retell+Gemini vs. migrate to OpenAI Realtime vs. migrate to Gemini Live vs. migrate to LiveKit-Agents-orchestrated stack

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0. Executive Summary

The pilot is at an unusual inflection: the **bridge, tools, and skills are ready** (393/393 unit tests, 7 voice tools, allergen Tier-3 handoff, recall_order, reservation lifecycle) — and the bottleneck has shifted from *application logic* to *speech I/O quality and tool-call reliability*. The Retell + Gemini 3.1 Flash Lite seam shows two structural problems that will not improve incrementally: (1) the cascaded **STT → LLM → TTS** architecture caps end-to-end latency at ~800–1300 ms TTFT regardless of LLM speed, and (2) function-call correctness on a non-realtime LLM driven through Retell's webhook layer is materially weaker than what speech-to-speech native models now deliver.

OpenAI's `gpt-realtime` (GA, Aug-Sep 2025) was specifically positioned as a **production voice-agent model**, scoring **66.5%** on ComplexFuncBench Audio (vs ~50% for the prior generation), with a 20% price cut, async function calling that no longer breaks turn-taking, native SIP, image input, and MCP tool support. Google's Gemini 2.5 Flash Live offers comparable feature breadth (30 HD voices, 24 languages, native barge-in, Affective Dialog) but the public signal in May 2026 still flags **production instability** — voice drift across weekly preview rotations, audio degradation past the ~1-minute mark, and missing first-class controls (speech rate, conversation toggles).

The recommendation embedded in the 2026-05-04 session resume — **OpenAI Realtime direct, LiveKit as a post-launch track** — survives this re-investigation. The asymmetry of risk favors migrating *before* commercial launch (TCR + Quantic + Maverick are all gated externally; we have free engineering cycles and only one pilot store to roll back if needed) rather than after, when fleet expansion makes regression cost compound.

Headline call: Migrate to OpenAI Realtime now (pre-TCR-approval window). Keep Retell as a fallback abstraction at the bridge boundary for ~30 days post-cutover. Defer Gemini Live until at least Q3 2026.

(한글 요약) 파일럿 상용화 직전인 지금 OpenAI Realtime으로 옮기는 것이 옳다. 현재 Retell+Gemini 3.1 Flash Lite는 STT→LLM→TTS 캐스케이드 구조상 TTFT가 800–1300ms 이하로 내려가지 않고, 함수 호출 정확도도 native speech-

to-speech 모델 대비 구조적 열위. Gemini Live는 기능은 비슷하지만 2026년 5월 시점에서 production 불안정성(음성 드리프트, 1분 이후 품질 저하, 핵심 컨트롤 부재)이 보고되고 있어 Q3 2026까지 대기 권장.

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1. Migration Timing Justification

1.1 The Pre-Commercial Window Is the Cheapest Migration Window You Will Ever Have

Three external dependencies are still pending: **Twilio TCR (10DLC) approval, Quantic POS white-label contract, Maverick payment integration spec**. All three block the actual go-live. None of them block code work. This produces a "free" engineering window with a measurable end date — exactly the situation where a foundational architecture swap is cheapest.

The asymmetry of "migrate now" vs "migrate later":

Dimension	Migrate now (pre-launch)	Migrate after launch
Active stores affected during cutover	1 (JM Cafe)	1 → N (Phase 1: 5 PDX stores)
Live-call regressions visible to customers	0 (only internal test calls)	Full call-volume blast radius
Pressure to roll back after a single bad call	Low (no SLA, no contract)	Very high (paying SMB owner)
Sales narrative cost	None	"We're rebuilding our voice stack" mid-deal
Engineering capacity competing with go-to-market	Low (TCR/Quantic/Maverick are blocking)	High (deployments, support tickets, training)
Test surface	393 unit tests, controlled	393 + production logs, partial coverage

1.2 What's Actually Wrong With Retell + Gemini 3.1 Flash Lite for Production

(a) TTFT floor is structural, not tunable. The current 800–1300 ms TTFT is the sum of: VAD endpointing (~150–250 ms) + Retell STT (~200–300 ms) + Gemini 3.1 Flash Lite first-token (~250–400 ms) + Retell TTS first-byte (~200–300 ms). No single component is "broken" — but the cascade itself ceilings at ~800 ms even on a perfect day. OpenAI Realtime documents typical end-of-user-speech to start-of-AI-audio of **300–500 ms**, with TTFB of ~500 ms from US clients. That is the *floor*, not the *ceiling*, of what an SMB caller perceives as "human."

(b) Function call accuracy on cascaded stacks is not where it needs to be. OpenAI's published benchmark — ComplexFuncBench Audio jumping from 49.7% (Dec 2024 model) to 66.5% (gpt-realtime) — is the most honest data point in this market. Our pilot has 7 tools, several with strict schemas

(`make_reservation` with party_size + datetime + name disambiguation, `modify_order` with variant_id resolution, `recall_order`). Errors at the JSON-arg level surface as silent failures (wrong reservation time, wrong drink modifier) — exactly the failure mode that destroys SMB owner trust on day 1.

(c) Interruption / barge-in on cascaded stacks is acceptable, not great. Retell's published latency (~780 ms response) is competitive in the cascaded-stack class but barge-in still requires the orchestrator to cancel a TTS stream that's already in flight. Native speech-to-speech models cancel in-model and resume context inside the same session, which is qualitatively different in messy environments (cafe background, kitchen noise — i.e. the literal target environment).

(d) Multilingual code-switching. PDX SMB demographics make Spanish-EN code-switching common, and JM Tech One's wedge thesis includes Korean. Cascaded stacks code-switch by routing — every switch round-trips through STT detection and TTS voice swap. gpt-realtime explicitly trained for "switching seamlessly between languages mid-sentence." For F&B verticals this is not a feature, it is a daily occurrence.

(e) Cost trajectory. Retell's true loaded cost (voice engine + LLM + telephony) is **$0.13–$0.31/min**. gpt-realtime audio is $32 / 1M input + $64 / 1M output tokens — roughly $0.06–$0.10/min for typical SMB call mixes after cached input savings, *before* telephony. Even granting a 30% wide error bar, the cost-per-minute compresses, not expands.

1.3 What's the Actual Risk of Migrating Now?

Honest list of failure modes:
1. **OpenAI Realtime SIP unreliability over Twilio.** OpenAI dev forum has open threads on SIP edge cases (calls ending unexpectedly mid-utterance). The community mitigation is the WebSocket-bridge pattern (Twilio Media Streams → custom WebSocket → OpenAI Realtime), which we already have the FastAPI bridge for. Net: the failure mode is known and the workaround is in our codebase shape.
2. **Voice quality regression on a specific phrase or accent.** Mitigation: keep Retell warm in shadow mode for 30 days post-cutover; the abstraction boundary at our FastAPI bridge already supports this.
3. **Token-cost surprise on long sessions.** OpenAI session cap is 60 min and now ships fine-grained context truncation. Mitigation: enforce conversation truncation at bridge layer.
4. **2-person team velocity loss.** Realistic — but the absence of competing deployment work during the TCR/Quantic/Maverick wait is the literal reason this is the cheapest window.

The risk of *not* migrating: shipping production with a TTFT and function-call ceiling we know we'll have to break later, after we have multiple paying SMB owners and zero rollback budget.

1.4 ROI in 2-Person + 1-Pilot Context

For a 2-person team, the dimension that matters most is **decision irreversibility**. Once JM Cafe is live and a second store is in pipeline, ripping out the speech I/O layer becomes a multi-week project with customer comms. Doing it now is a 1–2 week project with no comms. The migration itself is a forcing function for the abstraction we want anyway (a `VoiceTransport` interface above Retell/OpenAI/Gemini), which improves long-term flexibility regardless of which provider wins.

2. Four-Way Comparison Matrix (10-point scoring)

Each dimension scored 1–10 (10 = best in class for SMB voice agent, May 2026). Final row sums to a maximum of 90.

#	Dimension	Retell + Gemini 3.1 Flash Lite (current)	LiveKit Agents + OpenAI Realtime	OpenAI Realtime (direct)	Gemini 2.5 Flash Live
1	Speech naturalness, barge-in, multilingual EN/KR	6 (TTS competent, EN-only natural; KR mediocre; barge-in cascaded)	9 (gpt-realtime speech + LiveKit's adaptive interruption 1.5.x)	9 (gpt-realtime Cedar/Marin voices; mid-sentence language switching; native cancel)	7 (30 HD voices / 24 langs / barge-in good — but 1-min audio degradation, voice drift across previews reported)
2	Latency (TTFT, end-to-end, S2S vs cascade)	4 (800–1300 ms cascade; structural floor)	8 (~200–350 ms via S2S; LiveKit adds <50 ms orchestration)	9 (300–500 ms end-of-speech to AI-audio; 500 ms TTFB US)	7 (S2S architecture, but preview instability)
3	DX (SDK maturity, docs, debugging, function-call stability)	6 (Retell DX strong; Gemini function-calling on cascaded path is brittle)	8 (LiveKit Agents 1.5.x, MCP native, Python+Node, mature plugin ecosystem)	8 (Realtime SDK GA; Agents SDK; broad community; ComplexFuncBench 66.5%)	5 (Live API requires manual tool-response handling, no auto tool loop, declarations only at session start, "silent" tool execution unreliable)
4	Cost per minute (loaded)	5 ($0.13–$0.31/min loaded incl. telephony)	7 (~$0.08–$0.12/min audio + LiveKit Cloud or self-host)	8 ($32/$64 per 1M audio tokens; 20% price cut; cached input $0.40/1M)	8 ($0.30/$2.50 per 1M Flash; competitive but audio-token specifics opaque)
5	POS / SMB integration fit (function-call reliability + interrupt-safe context)	5 (works; modify_order variant_id resolution still fragile)	9 (LiveKit's tool-call orchestration + async tool exec patterns)	9 (async function calling no longer breaks session; MCP tool support; 66.5% ComplexFuncBench)	6 (manual tool-response loop adds bridge complexity; declarations frozen at session start limits dynamic skills)
6	Operational stability	7 (Retell mature; Gemini Lite	8 (LiveKit infra + OpenAI both	8 (GA, but SIP edge cases noted in dev	5 (still in preview/GA-edge

#	Dimension	Retell + Gemini 3.1 Flash Lite (current)	LiveKit Agents + OpenAI Realtime	OpenAI Realtime (direct)	Gemini 2.5 Flash Live
	(uptime, rate limit, regional)	reliable)	production-grade)	forum; WebSocket bridge mitigates)	for native audio; community reports voice drift week-over-week)
7	Data / privacy (ZDR, HIPAA, US SMB)	6 (Retell BAA available; Gemini Vertex enterprise tier yes; mixed surface)	8 (LiveKit self-host option + OpenAI ZDR/BAA)	8 (ZDR + BAA available on approval; need RBAC + audit logging)	7 (Vertex AI BAA path; audio-specific compliance docs less mature)
8	Lock-in risk	6 (two-vendor; LLM swappable; voice infra not)	9 (LiveKit is the abstraction — provider-swappable by design)	5 (single vendor; deep coupling to Realtime session model)	5 (single vendor; Vertex coupling)
9	Multilingual fit for PDX SMB (EN + ES + KR)	5 (EN strong; ES okay; KR weak on TTS naturalness)	8 (inherits gpt-realtime multilingual + LiveKit voice-provider swap)	8 (mid-sentence code-switching trained; 98 langs in training; KR pronunciation reasonable)	7 (24 HD voice langs incl. KR; ES strong; voice quality drift caveat)
Total	**/90**	**50**	**74**	**72**	**57**

Interpretation

- **LiveKit + OpenAI** scores highest on paper (74), driven by lock-in protection and DX. But it adds a framework dependency that the 2-person team must absorb during pre-launch — the same window that makes "now" cheap for the speech swap is the window where adding a new framework competes for attention. The 2026-05-04 decision to **defer LiveKit to post-launch track** stands.
- **OpenAI Realtime direct** (72) is 2 points behind LiveKit only on lock-in. Everything else is parity or stronger because there's no extra abstraction overhead during the migration.
- **Current Retell + Gemini 3.1 Flash Lite** (50) is below either OpenAI path by ~22 points — the gap is real, dominated by latency (4 vs 9), tool-call fit (5 vs 9), and speech naturalness (6 vs 9).
- **Gemini Live** (57) is currently a Q3-2026-revisit candidate, not a May-2026 migration target. Production instability + function-call ergonomics are the disqualifiers.

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3. OpenAI Realtime vs Gemini Live — Item-by-Item

3.1 Model lineup (May 2026)

	OpenAI	Gemini
Flagship	`gpt-realtime` (GA Aug-Sep 2025; refresh `gpt-realtime-mini-2025-12-15`)	`gemini-2.5-flash-live-api` ; `gemini-2.5-flash-preview-native-audio-dialog`
Mini variant	`gpt-realtime-mini` (cheaper, slightly lower instruction follow)	(Live tier rolls into 2.5 Flash)
Image input	Yes (gpt-realtime)	Yes (multimodal native)

3.2 Pricing (May 2026)

	OpenAI gpt-realtime	Gemini 2.5 Flash Live
Audio input	$32 / 1M tokens	Bundled in 2.5 Flash pricing — public docs imprecise on audio-only token rate
Audio output	$64 / 1M tokens	Same caveat
Cached input	$0.40 / 1M	Discount tier ~20% available
Text in/out reference	$0.30 / $2.50 per 1M (Flash baseline)	$0.30 / $2.50 per 1M Flash
Note	20% price cut vs prior gen; explicit audio-token line item	Audio token economics still less transparent in public docs as of 2026-05; production cost modeling requires Vertex pricing portal access

3.3 Latency benchmarks

	OpenAI gpt-realtime	Gemini 2.5 Flash Live
End-of-speech → AI audio start	300–500 ms typical	Comparable S2S range (preview); production stability less proven
TTFB (US client)	~500 ms	Comparable
Architecture	Native speech-to-speech, single multimodal model	Native audio output, but preview-tier voice drift reported

3.4 Function calling

	OpenAI gpt-realtime	Gemini 2.5 Flash Live
Mechanism	Function-calling events on side channel; async exec without breaking turn	Manual tool-response handling required (no auto loop)
Benchmark	ComplexFuncBench Audio: 66.5% (vs 49.7% prior)	No equivalent published voice tool-call benchmark

	OpenAI gpt-realtime	Gemini 2.5 Flash Live
Dynamic tool registration	Tools per session, updateable	All tools must be declared at session start — no mid-session add
Async tool execution	Native; long-running tools don't disrupt conversation	Async supported but "silent" execution unreliable; model may narrate tool exec
MCP support	Native MCP server tool calling	MCP available via plugins, less first-class in Live

For our 7-tool surface (create/modify/cancel order, make/modify/cancel reservation, allergen_lookup, recall_order), **OpenAI's session model is a closer fit**. The cancel/recall/modify trio especially benefits from async exec — these wrap Loyverse calls that occasionally take 2–5 s.

3.5 Interruption / barge-in / VAD

	OpenAI	Gemini
Barge-in	Native cancel on user speech (server-side VAD)	Native, "improved barge-in" advertised
Interruption sensitivity controls	Configurable VAD threshold; community reports it's tunable but undocumented edges	Some controls; speech-rate not first-class in Live

3.6 Transport

	OpenAI	Gemini
WebRTC	Yes (best perceived latency for browser/mobile)	Yes
WebSocket	Yes	Yes (default)
SIP	Yes (native SIP connector w/ Twilio Elastic SIP Trunking; some edge cases reported)	Via WebSocket bridge
Recommended for Twilio voice call	Twilio Media Streams → WebSocket bridge → OpenAI Realtime	Twilio Media Streams → WebSocket bridge → Gemini Live

3.7 SDK / DX

	OpenAI	Gemini
Python SDK	Mature; Realtime + Agents SDK	google-genai; less voice-specific affordance
Node SDK	Mature	Available
Browser	WebRTC sample apps in OpenAI cookbook	Examples repo on GitHub

	OpenAI	Gemini
Debugging	Side-channel events visible; community + cookbook substantial	Live API troubleshooting docs exist; community signal smaller

3.8 Voice options

	OpenAI	Gemini
Voice count	10 (alloy, ash, ballad, coral, echo, sage, shimmer, verse, **marin**, **cedar**)	30 HD voices
Recommended for production	Marin or Cedar (gpt-realtime tuned)	Voice quality drift reported on previews; testing needed per voice
Voice cloning	No	No (HD voice library only)

3.9 Context / session

	OpenAI	Gemini
Context window	32k (gpt-realtime) — 128k on some configs; instructions+tools cap 16,384 tokens	Larger Flash context (1M text equivalent), but Live session token limits less explicit
Max session	60 min (OpenAI direct); 30 min (Azure OpenAI)	Session caps less documented; reconnects common
Truncation	Auto at ~28,672 tokens; configurable	Manual context management more often required

3.10 Languages

	OpenAI	Gemini
Trained languages	98 (TTS + STT)	70 supported; 24 HD voice langs
Korean	Supported; mid-sentence switching trained	Supported; HD voice available
Spanish (PDX SMB priority)	Strong	Strong
Code-switching	Trained	Multilingual session support

3.11 Known constraints / drawbacks

OpenAI Realtime:
- SIP path has known edge cases over Twilio; WebSocket bridge is the safer production pattern.
- Voice cannot be changed mid-session once first audio response has been emitted.
- Cedar/Marin voices have a community-reported issue where they occasionally ignore agent system

instructions (GitHub issue openai/openai-agents-python#1746).
- Vendor lock-in is real — abstraction boundary must be designed deliberately.

Gemini Live:
- Voice drift across weekly preview rotations (Capella voice cited as broken).
- Audio quality degradation on TTS responses past ~1 minute.
- Speech rate not first-class; conversation behavior toggles missing.
- Function calling requires manual response handling; declarations frozen at session start.
- "Silent" tool execution unreliable — model may narrate execution unless guarded.
- Less production-tested in independent SMB voice deployments; preview/GA-edge.

3.12 Net call

For our pilot in May 2026, **OpenAI Realtime is the better match**. The deciding factors are: function-call benchmark transparency, async tool exec for Loyverse-bound tools, mature SDK + cookbook + community, and the absence of preview-tier instability reports. Gemini Live becomes a real second-source candidate **once native-audio preview stabilizes** (likely Q3 2026 by current cadence).

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4. Migration Procedure — Step-by-Step

4.1 Migration to OpenAI Realtime

Phase 0: Prerequisites (0.5 day)

- Provision OpenAI org with Realtime API access; request ZDR + BAA via Trust Center if any vertical (clinic, beauty) requires it.
- Confirm rate limits adequate for pilot (typically default tier covers single-store traffic).
- Network: outbound WSS to api.openai.com:443 from FastAPI bridge.
- Set env keys: `OPENAI_API_KEY`, `OPENAI_REALTIME_MODEL=gpt-realtime`.
- **Checklist**: API key live, BAA filed (if applicable), rate-limit dashboard visible.
- **Rollback**: revert env vars; bridge falls back to Retell path.

Phase 1: POC — Single-Call Echo (0.5 day)

- Standalone script: Twilio Media Streams → WebSocket bridge → OpenAI Realtime → echo reply.
- Verify: end-of-speech detection, audio out, end-to-end latency on real PSTN.
- **Risk**: codec mismatch (μ-law from Twilio vs PCM16 expected by OpenAI). Resample at bridge.
- **Time**: 4–8 hours.
- **Rollback**: discard branch.

Phase 2: Bridge integration — 7 tools (2–3 days)

- Map current 7 tools to Realtime function definitions:

- `create_order`, `modify_order`, `cancel_order`
- `make_reservation`, `modify_reservation`, `cancel_reservation`
- `allergen_lookup` (Tier-3 EpiPen handoff path preserved)
- `recall_order`
- Wire async tool executor (asyncio task, returns `function_call_output` event when done).
- Preserve our Bridge's existing `tenant_id` / RLS context propagation.
- **Risk**: schema strictness — Realtime is stricter than Gemini on JSON. Add JSON-schema validation at tool boundary.
- **Test**: invoke each tool 10× via voice; record JSON-arg correctness rate.
- **Rollback**: feature flag `VOICE_PROVIDER=retell|openai`; flip back at any time.

Phase 3: System prompt port (1 day)

- Port rule 1–13 + INVARIANTS from current Gemini system prompt.
- Re-tune for gpt-realtime instruction-following style (it follows literal style instructions like "speak empathetically" — exploit this).
- Encode allergen Tier-3 mandatory handoff as a hard rule (no probabilistic phrasing).
- Validate Korean greeting prompt naturalness with Marin and Cedar voices.
- **Risk**: instruction-follow regression on edge cases.
- **Test**: 30 scripted scenarios from existing test bank.

Phase 4: Pay link integration (0.5 day)

- Current: Retell post-call hook fires Twilio SMS pay link.
- Realtime equivalent: bridge listens for `response.done` + tool-call boundary, then triggers pay link.
- Preserve idempotency keys.
- **Risk**: hook timing — fire after tool completion, not after response.
- **Rollback**: keep Retell hook handler dormant.

Phase 5: Regression test (1 day)

- Run full 393/393 unit suite (no change expected — bridge tests are provider-abstracted).
- Add Realtime-specific tests: (1) tool-call schema, (2) async tool exec, (3) interruption mid-tool, (4) language switch mid-session.
- Live-call shadow: call JM Cafe pilot number 20× across morning/lunch/evening, score each call on 5 axes (latency, tool accuracy, speech naturalness, interruption handling, language).
- Target: ≥18/20 calls score "as good or better" than current stack.

Phase 6: Canary cutover (1 day live + 30 days monitoring)

- Day 0: 10% traffic to Realtime via feature flag (rollout key on caller hash).
- Day 1: 50%.
- Day 3: 100%.
- Retell warm in shadow mode (calls also routed for comparison logging, not to caller) for 30 days.

- **Rollback trigger**: if 2+ calls/day score below baseline on tool accuracy → flip flag back, RCA before retry.

Phase 7: Retell removal (1 day, day-30+)

- Remove Retell SDK from `requirements.txt`.
- Archive Retell webhook handlers under `app/adapters/_archive/`.
- Update `docs/architecture/ARCHITECTURE.md`.
- Keep `VoiceTransport` interface — this is the LiveKit-future-readiness investment.

Total estimated effort: 6–8 engineering days + 30 days monitoring.

4.2 Migration to Gemini Live (alternative path)

(Not recommended for May 2026; documented for completeness.)

Phase 0: Prerequisites (0.5 day)

- Vertex AI project with Gemini 2.5 Flash Live API access; or Google AI Studio API key (for non-enterprise).
- BAA via GCP if vertical requires.
- Set env: `GOOGLE_API_KEY` or service-account creds; `GEMINI_LIVE_MODEL=gemini-2.5-flash-live-api`.

Phase 1: POC echo (0.5–1 day)

- Same Twilio Media Streams → WebSocket bridge pattern.
- Audio chunk size: 20–40 ms (Gemini Live constraint).
- **Risk**: chunk-size discipline — wrong chunking visibly increases latency.

Phase 2: Tool integration (3–5 days)

- All 7 tools declared at session start (Live API limitation — no mid-session add).
- **Manual tool-response handling**: write our own loop that listens for `tool_call`, executes, and posts `tool_response`. No auto-orchestration like OpenAI provides.
- Add explicit "silent execution" guardrails in system prompt to suppress narration of tool exec — and/or use fire-and-forget pattern (don't post FunctionResponse) for async-style tools.
- **Risk**: silent execution leakage where model verbally narrates a tool call (documented Live API behavior).

Phase 3: System prompt port (1–2 days)

- Re-tune for Gemini Live native-audio behavior; voice drift means production validation per HD voice.
- Affective Dialog tuning available — use for empathetic allergen Tier-3 handoff.

Phase 4: Pay link integration (0.5 day)

- Bridge-driven, same pattern as OpenAI path.

Phase 5: Regression test (1.5 days — heavier than OpenAI due to manual tool loop validation)

- Same 393/393 unit suite.
- Additional: voice-quality regression test across HD voices we use (drift risk).
- 1-minute-plus TTS quality check (known degradation point).

Phase 6: Canary cutover (1 day + 30 day monitoring)

- Same canary pattern. Higher rollback probability given preview-tier instability.

Phase 7: Retell removal (1 day)

- Same as OpenAI path.

Total estimated effort: 8–11 engineering days + 30 days monitoring (plus higher monitoring intensity due to preview-tier risk).

4.3 Side-by-side migration cost comparison

Phase	OpenAI Realtime	Gemini 2.5 Flash Live
Prereqs	0.5 d	0.5 d
POC echo	0.5 d	0.5–1 d
Tool integration	2–3 d	3–5 d (manual tool loop overhead)
Prompt port	1 d	1–2 d (per-voice tuning)
Pay link	0.5 d	0.5 d
Regression	1 d	1.5 d
Canary	1 d + 30 d watch	1 d + 30 d watch (higher intensity)
Retell removal	1 d	1 d
Total dev	**6–8 days**	**8–11 days**

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5. Risk Register and Mitigations

Risk	Severity	Likelihood	Mitigation
OpenAI Realtime SIP edge cases over Twilio	Medium	Medium	Use WebSocket-bridge pattern (already in our codebase shape), not native SIP connector

Risk	Severity	Likelihood	Mitigation
Voice quality regression on a corner-case phrase or accent	Medium	Low-Medium	30-day Retell shadow mode; per-call scoring rubric
Tool-call JSON schema strictness vs current looser handling	Low-Medium	Medium	Add JSON-schema validator at tool boundary; expand schema test fixtures
Token-cost surprise on long sessions	Low	Low	Enforce conversation truncation in bridge; cap session at 30 min for single-call use case
Cedar/Marin voice ignoring system instructions (open issue)	Low	Low	Use `verse` or `coral` as fallback voice; pin voice version once chosen
Two-person team velocity hit during pre-launch	Medium	Medium	Migration is sequential with TCR/Quantic/Maverick blockers — net cost of work is near zero
Vendor lock-in on OpenAI	Medium	High (it is real)	Maintain `VoiceTransport` interface; LiveKit becomes the post-launch hedge

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6. Recommendation Summary

1. **Migrate to OpenAI Realtime now**, during the TCR/Quantic/Maverick external-blocker window.
2. **Estimated cost**: 6–8 engineering days + 30-day shadow-mode monitoring.
3. **Keep Retell warm for 30 days post-cutover** as a hard rollback path.
4. **Defer Gemini Live to Q3 2026 re-evaluation** when native-audio preview is expected to stabilize.
5. **Defer LiveKit Agents to post-launch** as the lock-in hedge — but design the `VoiceTransport` interface now so LiveKit insertion is a 1–2 day swap later, not a rebuild.
6. **Voice choice**: start with `marin` for general F&B, validate `cedar` for premium-tone vertical (clinic/beauty), keep `verse` as fallback.
7. **Function calling**: implement async-tool-exec pattern from day 1 — Loyverse calls regularly take 2–5 s and we must not block turn-taking.

(한글 요약) 즉시 OpenAI Realtime 마이그레이션 권장. 6–8 엔지니어링 일 + 30일 shadow 모니터링. Retell은 30일 fallback로 유지. Gemini Live는 Q3 2026 재평가. LiveKit은 launch 이후 lock-in hedge로 추가. 음성은 marin 시작, 비상시 verse fallback. 7개 툴 모두 async tool exec 패턴 적용.

Sources

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4. Realtime API with SIP — OpenAI — Native SIP support.
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End of report.